

1. Administrative & Study Identification

Full Study Title: An Open-label, Randomized Phase 3 Study of MK-2870 as a Single Agent and in Combination With Pembrolizumab Versus Treatment of Physician's Choice in Participants With HR+/HER2- Unresectable Locally Advanced or Metastatic Breast Cancer
Short Title/Acronym: TroFuse-010 (MK-2870-010) **Protocol Number:** MK-2870-010 **NCT Number:** NCT06312176 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Sacituzumab Tirumotecan (MK-2870) and Pembrolizumab (MK-3475) **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead **Study Status:** Recruiting

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether sacituzumab tirumotecan as a single agent and sacituzumab tirumotecan plus pembrolizumab are superior to Treatment of Physician's Choice (TPC) with respect to progression-free survival (PFS) in all participants. **Secondary Objectives:** To evaluate Overall Survival (OS), Progression-Free Survival (PFS) comparing pembrolizumab plus sacituzumab tirumotecan versus sacituzumab tirumotecan alone, Objective Response Rate (ORR), Duration of Response (DOR), patient-reported outcomes (PROs), and safety.

2.2 Background & Rationale Additional therapies are needed to improve outcomes in patients with HR+/HER2- breast cancer that progressed on endocrine therapy (ET) plus CDK4/6 inhibitors (CDK4/6i). Trophoblast cell surface antigen 2 (TROP2) is commonly overexpressed in patients with HR+/HER2- metastatic breast cancer (mBC) and is associated with a poor prognosis. This study investigates whether MK-2870 (a TROP2-directed antibody-drug conjugate), alone or in combination with pembrolizumab (an immune checkpoint inhibitor), provides superior clinical benefit compared to standard chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Treatment of Physician's Choice - TPC)
Blinding: Open-label **Randomization:** Randomized (3:3:2 ratio)
Disease Area: Breast Neoplasms

3.2 Planned Enrollment & Duration Target **\$N\$:** ~1,200 participants
Number of Sites: Global multicenter study **Estimated Study Start:** 13-Apr-2024 **Estimated Primary Completion:** 11-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before randomization. 2. Has unresectable locally advanced or metastatic centrally-confirmed hormone receptor positive (HR+)/human epidermal growth factor receptor 2 negative (HER2-) breast cancer. 3. Has radiographic disease progression on one or more lines of endocrine therapy for unresectable locally advanced/metastatic HR+/HER2- breast cancer, with one in combination with a CDK4/6 inhibitor. 4. Is a chemotherapy candidate and has adequate organ function. 5. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on antiretroviral therapy. 6. Participants who are Hepatitis B surface antigen (HBsAg) positive are eligible if they have received HBV antiviral therapy for at least 4 weeks, and have undetectable HBV viral load. 7. Participants with a history of Hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has breast cancer amenable to treatment with curative intent. 2. Has received prior chemotherapy for unresectable locally advanced or metastatic breast cancer. 3. Has experienced an early recurrence (<6 months after completing adjuvant/neoadjuvant chemotherapy) and is therefore eligible to receive second-line (2L) treatment. 4. Has symptomatic advanced/metastatic visceral spread at risk of rapidly evolving into life-threatening complications. 5. Active autoimmune disease that has required systemic treatment in the past 2 years. 6. History of (noninfectious) pneumonitis/interstitial lung disease that requires steroids, or has current pneumonitis/interstitial lung disease. 7. Has an active infection requiring systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sacituzumab Tirumotecan (MK-2870) administered via IV infusion every 2 weeks. Pembrolizumab administered via IV infusion every 6 weeks. Treatment of Physician's Choice (TPC) involves standard chemotherapy (e.g., capecitabine, nab-paclitaxel, paclitaxel) dosed per standard of care. **Route of Administration:** Intravenous (IV) and Oral (depending on TPC assignment). **Duration of Treatment:** Until disease progression, intolerable toxicity, or study withdrawal (up to approximately 2 years for pembrolizumab).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Concurrent

investigational agents, live vaccines, and other systemic anti-cancer therapies not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Central Biomarker Assessment (TROP2, HR, HER2, PD-L1)
Baseline	Day 0	Randomization, First Dose
Interim	Q9W to Week 54	Treatment administration, Safety Labs, Tumor imaging (Every 9 weeks)
End of Study	Week 54+	Tumor imaging (Every 12 weeks after week 54), Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free survival (PFS) in all participants, defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause. Assessed for sacituzumab tirumotecan versus TPC, and pembrolizumab + sacituzumab tirumotecan versus TPC. PD is defined as $\geq 20\%$ increase in the sum of diameters of target lesions with an absolute increase of ≥ 5 mm, or the appearance of one or more new lesions. **Measurement Method:** Assessed by blinded independent central review (BICR) per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1).

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Progression-Free Survival (PFS) (pembrolizumab + sacituzumab tirumotecan versus sacituzumab tirumotecan) - Objective Response Rate (ORR) - Duration of Response (DOR) - Patient-Reported Outcomes (PROs) and Safety profile

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via scheduled onsite visits, laboratory testing, and physical examinations. **Serious Adverse Events (SAEs):** Must be reported within standard 24-hour regulatory timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to evaluate safety and efficacy data across study arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes (PFS). Safety Set for all participants who receive at least one dose of study treatment. **Sample Size Justification:** Target enrollment of $N = 1,200$ is adequately

powered to detect a statistically significant superiority in PFS for both experimental arms compared to TPC. **Handling of Missing Data:** Standard censoring rules for time-to-event analyses using Kaplan-Meier survival curves.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Comprehensive onsite and remote clinical monitoring by the Sponsor to ensure protocol adherence and patient safety. **Audit Trail:** Secure Electronic Data Capture (EDC) system tracking with complete audit trails and centralized radiological review.

11. Study Identifiers & Governance

Primary ID: MK-2870-010 **Registry Number:** NCT06312176 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** TroFuse-010 **Official Regulatory Title:** An Open-label, Randomized Phase 3 Study of MK-2870 as a Single Agent and in Combination With Pembrolizumab Versus Treatment of Physician's Choice in Participants With HR+/HER2- Unresectable Locally Advanced or Metastatic Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Unresectable Locally Advanced or Metastatic HR+/HER2- Breast Cancer (Breast Neoplasms) **Diagnosis Threshold:**

Centrally-confirmed HR+/HER2- status with radiographic disease progression on prior endocrine therapy **Medical Methodology:**

TROP2-directed Antibody-Drug Conjugate (ADC) combined with Immune Checkpoint Inhibitor (PD-1) vs. standard chemotherapy

Optimization Target: Disease control, Objective Tumor Response, and extended Progression-Free Survival

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced metastatic disease

Education Protocol (HCP): Protocol-specific training on ADC safety, infusion reactions, and immunology management

Education Protocol (Patient): Informed consent detailing trial expectations, treatment schedules, and self-monitoring of potential AEs

Quality Audit Mechanism:

Centralized RECIST 1.1 evaluations (BICR) to prevent investigator bias

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression, death, or study withdrawal (Up to ~7 years) **Onsite Visit Cadence:** Every 2 weeks for MK-2870 dosing and safety checks; Every 6 weeks for Pembrolizumab **Remote Monitoring Frequency:** As indicated for safety reporting and survival follow-up **Checklist Review Interval:** Tumor imaging every 9 weeks (Q9W) through Week 54, then every 12 weeks (Q12W)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				